

Motor Datasheet: 2026-08-19

17-31-57 output

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	34.1	mm
Machine Inner Diameter	22.8	mm
Machine Thickness / Width	17.0	mm
Stator Height	17.0	mm
Magnet Height	2.2	mm
Tooth Height	8.1	mm
Air Gap	0.50	mm
Fill Factor (input)	0.40	-
Slot Fill Factor (calculated)	Not calculated	-
Slot Width (tooth to tooth)	5.096	mm
Available Coil Space	2.548	mm
Assumed Coil Width (simulation)	2.348	mm
Assumed Coil Height (simulation)	7.912	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	21.6	g
Rotor Mass	8.3	g
Stator Mass	13.4	g
Total Copper Mass (pure copper)	11.2	g
Total Stator Steel Mass	2.1	g

Total Magnet Mass	2.3	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	16.8	g·cm ²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	1.797	A
Supply Current (Peak)	2.541	A
Id (Peak)	-0.000	A
Iq (Peak)	2.541	A
Phase Advance Angle	0.0	°elec
Phase Resistance at operational temperature	1063.222	mOhm
D-Axis Inductance (Ld)	513.908	μH
Q-Axis Inductance (Lq)	466.254	μH
Terminal Inductance (Ltt = Ld + Lq)	980.162	μH
PM Flux Linkage	1.272	mWb
Frequency at operating point (electrical)	2858.333	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	48.000	V
Power Factor	0.775	-

Performance

Parameter	Value	Unit
Torque (Continuous)	30.7	mNm
Speed	24500.0	rpm
Dimensionless motor constant	3.3	-
BackEMF Constant (Ke, phase peak)	0.9	mV/rpm
Motor Constant (Km, only resistive losses)	9.6	mNm/√W
Specific Motor Constant (Km / mass)	442.6	mNm/√W/kg
Torque Constant (Kt, peak, no-load)	13.8	mNm/A

Torque Ripple	3.0	%
Saturation	12.5	%

Losses

Parameter	Value	Unit
Total Losses	13.8	W
Resistive Losses	10.3	W
Iron Core Losses	2.3	W
Magnet Eddy Current Losses	0.1	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

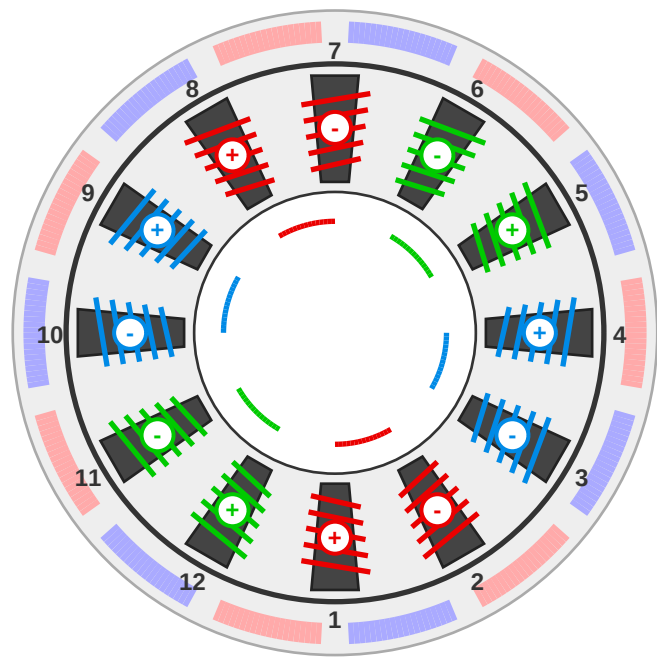
Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	12	-
Pole Pairs	7	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	91.0	-
Number of Phase Turns (all phases in series)	364.0	-

Winding Configuration

Schema: **AabBCcaABbcC**

Concentrated Winding | Balanced | Cogging: 84/rev | Ripple: 84/rev



Materials

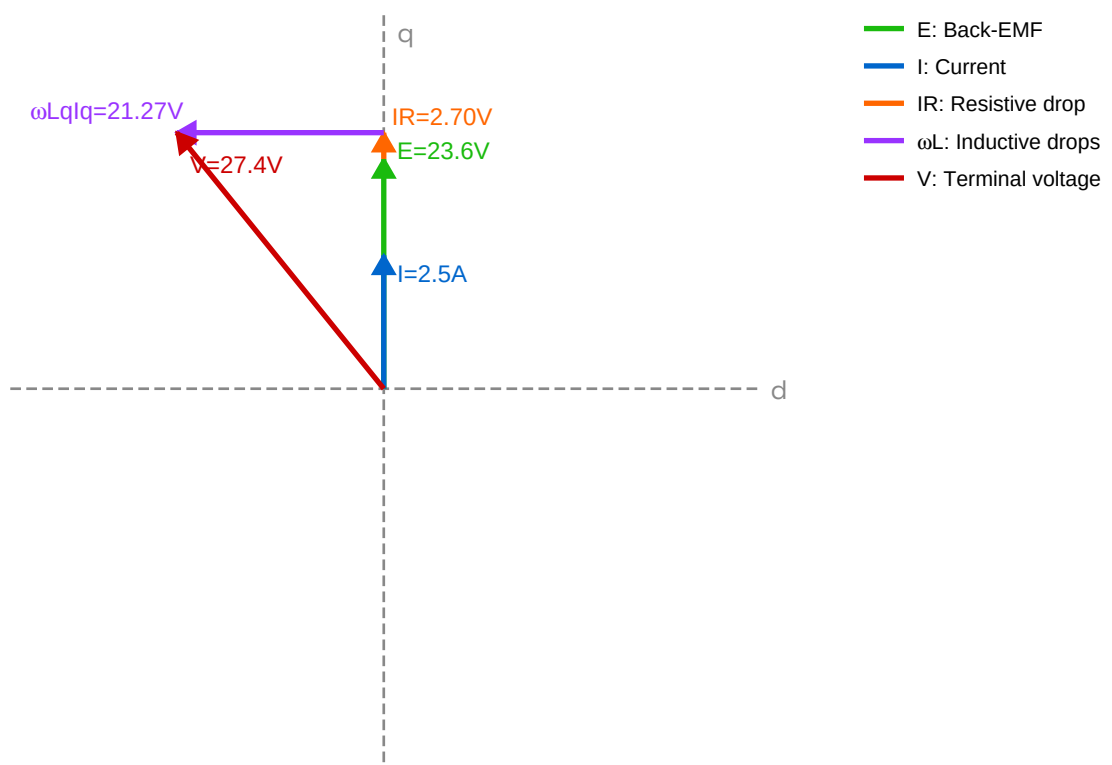
Component	Material	
Stator Core	N030-Cogent	-
Magnets	BMN-52SH (GBD)	-
Rotor Backiron	11SMn30	-
Coil	Copper	-

Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	23.614	V
Resistive Drop (Phase, Peak)	2.701	V
Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	21.275	V

Inductive Drop Total (Peak)	21.275	V
Terminal Voltage (Phase, Peak)	27.419	V
Back-EMF Voltage (Line-to-Line, Peak)	40.901	V
Resistive Drop (Line-to-Line, Peak)	4.679	V
Inductive Drop (Line-to-Line, Peak)	36.849	V
Terminal Voltage (Line-to-Line, Peak)	47.490	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



Mechanical Properties

Parameter	Value	Unit
Airgap Axial Force (Average)	0.22	N
Magnet Axial Force Positive Maximum	3.46	N
Magnet Axial Force Positive Minimum	1.97	N
Magnet Axial Force Negative Maximum	0.00	N
Magnet Axial Force Negative Minimum	0.00	N

Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms: $\text{Peak} = \text{RMS} \times \sqrt{2}$.
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection: $V_{LL} = V_{\text{phase}} \times \sqrt{3}$.